A Guided Deep Learning Approach for Joint Road Extraction and Intersection Detection From RS Images and Taxi Trajectories

Yali Li, Longgang Xiang, Caili Zhang, Fengwei Jiao, and Chenhao Wu

***Abstract*—Automatic extraction of road information based on data-driven methods is significant for various practical applications. Remote sensing (RS) images and GPS trajectories are two available data sources that can describe roads from a complementary perspective, and fusing them together can improve road detection performance. However, existing studies on the combination of RS images and GPS trajectories do not fully utilize their enhanced information about roads and suffer from road information loss. Moreover, roads and intersections are two crucial elements of road network generation, and they are closely related to each other. Therefore, we propose a multitask and multisource adaptive fusion (MTMSAF) network, which leverages RS images and trajectory datasets to execute road extraction tasks and intersection detection tasks simultaneously. The MTMSAF network is built on three components. First, two encoder steams are designed to capture road features from RS images and trajectories. Then, an adaptive fusion module is created to fuse the individual road features at each level in a guiding fashion. Finally, two specific decoders are proposed: one is used to implement the road region detection task by considering multidirectional geometric road information, and the other is used to perform the intersection extraction task by recovering intersection information with the road region feature. Four state-of-the-art deep learning-based methods, including segmentation networks and road extraction networks, are compared with the proposed approach. The results show the superiority of our approach for both road detection and intersection extraction tasks.**

***Index Terms*—GPS taxi trajectories, guidance strategy, intersection extraction, remote sensing (RS) images, road detection.**

I. Introduction

COMPLETE and accurate road networks, which consist of road segments and intersections connected in series, are essential for practical applications such as intelligent transportation [1], updating geographic information, and urban planning [2]. With the acceleration of urban construction, road patterns also change at high speeds. Road network generation using traditional methods such as surveying possesses a long time requirement, high cost, and small coverage, so it cannot meet the demand for road map construction and updating. To this end, many scholars have focused on developing data-driven approaches to extract road information automatically, such as GPS trajectory-based methods [3]–[5], remote sensing (RS) image-based methods [6]–[8], and laser point cloud-based methods [9].

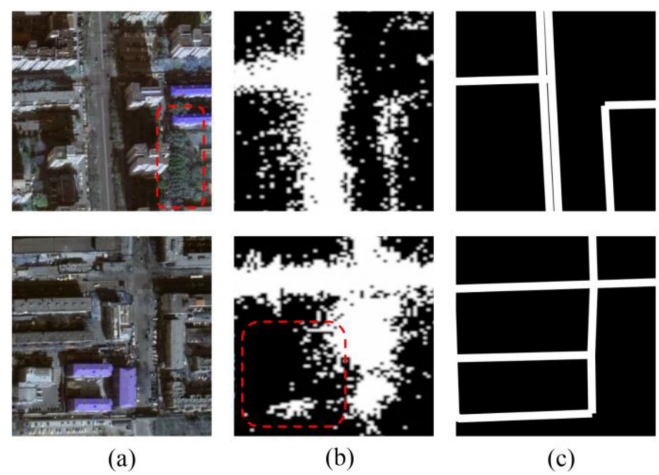

Fig. 1. Example showing complementary descriptions of roads from RS images and taxi trajectories. (a) RS images. (b) Trajectories. (c) Ground truth.

RS images and vehicle trajectories are two main reliable data sources for road information extraction, with advantages such as low costs, regular updates and wide coverage [10], [11]. However, roads are usually obscured by occlusion, and road-like objects exist in RS images, making it challenging to extract complete and accurate roads. The drawbacks of taxi trajectories, such as noise [12], low frequencies, and uneven distributions [13] also lead to the decline in integrity and quality of road extraction results. Overall, it is still a challenge to acquire complete and accurate road information based only on RS images or taxi trajectories. Nevertheless, RS images and taxi trajectories are complementary in road description [14], as shown in Fig. 1. On the one hand, a road may be invisible or confused with road-like objects in the RS image but clearly displayed in the trajectories, as shown in the red box in the first row of Fig. 1. On the other hand, roads that are covered by small track points or under the interference of noise may be difficult to identify in the trajectories but easy to distinguish in the RS

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images, as shown in the red box in the second row of Fig. 1. Therefore, it is necessary to integrate RS and trajectories in road information extraction and find an efficient fusion method to combine their respective advantages and improve the quality of road extraction. Although there are some preliminary studies on road extraction based on fused RS images and trajectories, the technology of data fusion is relatively simple, such as the concatenation of two data sources directly as the input of the deep learning network [14], [15]. The lasted fusion method is DeepDualMapper [16], which designed a gated module to learn a complementary-style fusion.

Roads and intersections are two basic elements of road networks. The existing methods treat road extraction and intersection detection separately and adapt two-step approaches, which include road extraction first and then intersection separation [17], or intersection detection first and then fitting the road that are between intersection pairs [18]. However, to a certain extent, road extraction and intersection detection depend on each other. The road detection result contains intersections, since it is easy to obtain intersections from road detection by using tensor voting [19] or by analyzing the connectivity of the morphological thinning result of the road [15]. Meanwhile, the intersections can help to link broken roads around junctions. Furthermore, two-step methods are highly dependent on the intermediate results, and errors in the first stage are transmitted to the later information extraction stage. Therefore, it is necessary to couple road detection and intersection extraction in a one-step method to improve their joint performances.

Deep learning based methods have been applied to road information identification due to their strong abilities of feature learning and expression. Most deep learning models were developed to solve one task of road extraction based on state-of-the-art networks and their variants. In addition, instead of considering road information extraction as a single task, some scholars also took various road elements into account that were closely related and incorporated a multitask deep learning network into road identification. Cheng *et al*. produced a cascaded convolutional neural network [18]. Lu *et al.* proposed a multiscale and multitask deep learning framework [20], Liu et al. proposed a RoadNet [21] and Yang *et al.* introduced a recurrent convolutional neural network known as U-Net [22]. The networks mentioned above are all used for performing road detection and road centerline extraction from RS images simultaneously, and they improve the extraction precision of the two tasks. However, to the best of our knowledge, there are no previous studies on predicting roads and intersections simultaneously based on deep learning technologies.

Given the problems in the existing works, we propose a multitask deep learning approach to facilitate road detection and road intersection extraction tasks simultaneously by fusing RS images and taxi GPS trajectories, which is named the multitask and multisource adaptive fusion (MTMSAF) network. We build the MTMSAF network based on a popular encoder-decoder structure to capture the contextual information at multiple scales. In the encoder, we first extract road features from RS images and taxi GPS trajectories separately, and then the guidance mechanism is introduced to fuse the road features from two data sources at each level. Moreover, instead of the isolated extraction of the two-stage framework used in previous studies, the MTMSAF network employs a multitask learning scheme, which adopts the road decoder and intersection decoder to resolve road extraction and intersection detection tasks simultaneously. In the proposed multitask learning framework, the domain knowledge learned from one task can be directly utilized in the other task, which leads to more consistent and better performance. In summary, our main contributions are as follows.

1) A novel encoder-decoder-based deep learning network is designed with an integrated multitask learning scheme to solve road detection and intersection extraction tasks simultaneously based on the fusion of RS image features and trajectory features. Thus, better detection and more complete performances are achieved.
2) An adapted fusion module is designed with a guidance strategy, which fuses road features captured from RS imagery and trajectories at each level. In this way, road enhancement information can be better exploited and good road continuity can be obtained.
3) Two individual decoder branches are applied to couple road detection and intersection detection tasks. One is the road decoder branch, which is designed with the goal of measuring the continuity of roads from different directions to alleviate gaps. The other is the intersection decoder branch, which is developed to highlight potential junction areas through the guidance strategy.

The rest of this article is organized as follows. Section II presents related works. In Section III, we first describe the architecture of the framework and then explain the designed encoder and decoder in detail. Section IV presents the experimental results and a comparative analysis. Finally, Section V discusses conclusion and future studies.

II. Related Works

Road information extraction via RS image is a hot research topic. Traditionally, processing of RS image data is using efficient exposure features, texture characteristics, and so on to obtain the target information through specific methods, such as wavelet transform [23], [24], In recent years, inspired by the performance of state-of-the-art deep learning technology in the computer vision community, such as image classification, object detection [25], [26], and image segmentation [27], deep learning has been also incorporated into RS image processing and obtained appealing results. Road information extraction is a research hotpot in RS field. Mnih and Hinton [28] first applied a deep learning network called a restricted Boltzmann machine. However, their framework is less effective than a CNN [11] in road segmentation from RS images. Inspired by the spatial accuracy of U-Net [29], many scholars utilized U-Net to achieve promising performances in road extraction [30]. However, the continuity of the road segments is not guaranteed. Recently, Cheng *et al.* [31] combined directional information to facilitate continuous road extraction. Tao *et al.* [32] proposed a road-specific information inference structure to preserve the continuity of the extracted road. However, these methods are

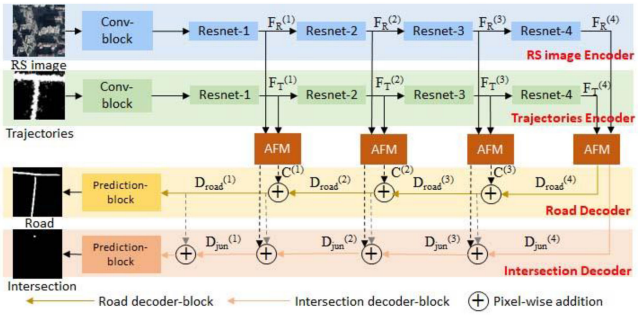

Fig. 2. MTMSAF network architecture.

based on contextual information and road trends, and thus, their road reasoning ability is reduced in environments that lack such contextual information, such as tree-lined avenues, or environments where road trends are variable, such as intersections.

Road information extraction from vehicle trajectories has been well studied. The methods can be divided into point-based, line-based, grid-based, and intersection-connection methods. The point-based methods obtain the characteristic points based on clustering algorithms [33], [34], where clusters represent either junction points or road section points. Line-based methods select one line of vector trajectories as initial objects, using the gravity model [35] and weighted Delaunay triangle model [36] to gradually add and optimize the positions of the new track lines in the road network. Grid-based methods rasterize the scatter trajectory points and apply mathematical morphology [37] and topical model-based methods [38] to extract the road centerlines. Intersection-connection methods extract road intersections by calculating trajectory sampling point characteristics [39], such as density, direction, and speed, or by analyzing the geometric features of the road [40], and then generate road links to connect the intersections. However, these methods still cannot eliminate noise or extract roads in regions with few or even zero tracking points, such as low-level roads.

Employing only a single data source limits the performance of road extraction, and some researchers have worked on using both GPS trajectory data and RS images to facilitate road extraction. One method is to extract roads from one data source based on the processing results of the other data source [41]. The other methods concatenate RS images and GPS trajectory feature maps as the inputs of deep neural networks [14], [15]. These fusion methods are not sufficiently delicate to fully mine the enhanced information between two data sources, nor do they consider obtaining various road information simultaneously based on fusion data.

III. Method

In this section, we explain the proposed deep learning approach for road detection and intersection extraction in detail, which is named the multitask and multisource adaptive fusion (MTMSAF) network. The architecture of the proposed approach is illustrated in Fig. 2, and three components are involved in the architecture. 1) Two streams are utilized to capture road information from RS images and GPS trajectories. 2) The adaptive fusion module (AFM) is designed to fuse road features learned from two data sources. 3) Two decoder branches are introduced to facilitate the road extraction task and intersection detection task.

A. Multisource Stream Encoders

The challenge in recognizing road information from RS images and trajectories is that the appearance of the same road is quite different in RS images and trajectory feature maps. To this end, instead of concatenating two types of data directly as inputs like in most previous works, we utilize two streams to learn road features from RS images and trajectories.

The two independent streams possess the same network structure, which uses pretrained ResNet34 [42] as the encoders. Considering the difficulty in capturing road features from the comprehensive context of RS images, we utilize the parameters of the pretrained D-LinkNet module [43], which achieves superior performance in road extraction, to initialize the weights of the RS image extraction encoder. In this way, the multilevel and multitype road features are obtained from two streams. The output features of the *i*th residual learning block named $F_R^{(i)}$ and $F_T^{(i)}$ represent the *i*th extracted feature from the RS image and the *i*th extracted feature from the trajectories, respectively, i.e., $F_R^{(1)} \sim F_R^{(5)}$ and $F_T^{(1)} \sim F_T^{(5)}$ in Fig. 2.

B. Multisource Feature Adaptive Fusion Module

Two types of multiscale activation maps are extracted from RS images and trajectory data based on the two-stream structure, yet how to utilize the advantages of the two kinds of road features and make them complement one another is still unresolved. To solve this problem, the AFM is proposed.

The motivation of our AFM lies in the fact that RS images and trajectories are two data sources describing road information from different viewpoints and providing enhanced information with respect to roads that is potentially helpful for improving road continuity and completeness. Thus, if we can emphasize some areas in the road feature map obtained from one data source according to the road information extracted from another data source through the guidance strategy and combine the mutual guidance results, we can then obtain the comprehensive road fusion information. To this end, we build a feature fusion block (FFB) to capture the enhanced information and fuse the guidance result through a concatenation way.

The FFB is motivated by the pixel-adaptive convolution (PaConv) [44], which can adaptively emphasize the pixel weights according to the guiding feature and is different from convolutions that share spatial weights. PaConvs are selected to build the FFB to make a semantic fusion of guidance features and inputs through the guidance method. The guidance method of the PaConv is to modify standard filter weights by multiplying by a variant kernel $K \in \mathbb{R}^{c' \times c \times s \times s}$, which depends on the guidance features f. For the image features $x = (x_1, x_2, \ldots, x_n)$, $x_i \in \mathbb{R}^c$ over *n* pixels and c channels, as well as the c'-channel

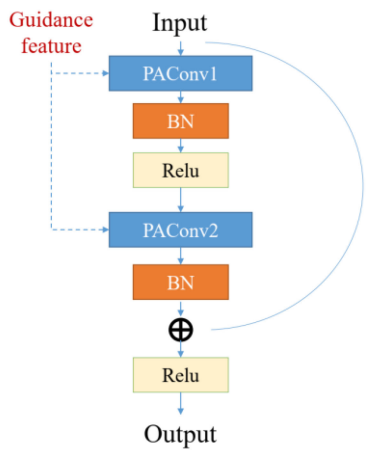

Fig. 3. FFB architecture.

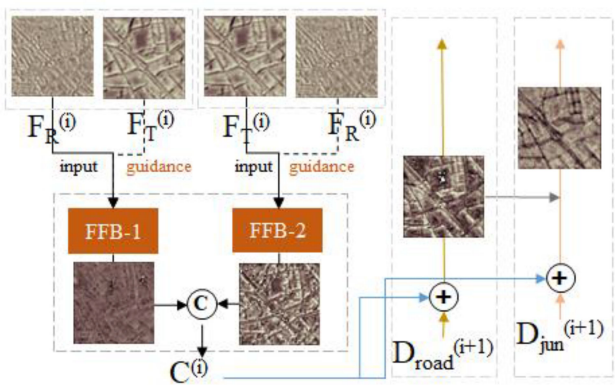

Fig. 4. Demonstration of the AFM.

output, the PaConv can be defined as (1)

$$x_i' = \sum_{j \in W(i)} K\left(f_i, f_j\right) W_j x_j + b \tag{1}$$

where $W(i)$ is defined as an $s \times s$ sliding convolution window, $W = (W_1, W_2, \ldots, W_{s\times s})$, $W_j \in \mathbb{R}^{c' \times c}$ are the filter weights, and $b \in \mathbb{R}^{c'}$ is the bias. K is a Gaussian kernel function $\mathrm{K}(p_i, p_j) = \exp\{-\frac{1}{2}(p_i - p_j)^\top (p_i - p_j)\}$. The spatially invariant filter W is modified at distinct pixels by the calculation of f through K, and the guidance feature f is either predefined or learned from the deep learning network.

Considering the advantages of residual learning in exploiting rich spatial context and alleviating vanishing gradients and accuracy degradation, residual learning is incorporated into the construction of the FFB. In this way, the input road feature is used together with the guided road feature result at the same time. Different from the traditional residual block, FFB employs two 3×3 PaConvs instead of two regular convolutions, and except for the input, the guidance feature is utilized to perform the two PaConvs. The structure of the FFB is shown in Fig. 3.

After defining the FFB, the AFM is proposed, as shown in Fig. 4, which includes two FFBs and one concatenation operation. In detail, the outputs of each level from the two streams (i.e., $F_R{}^{(i)}$ and $F_T{}^{(i)}$) are treated as either input or guidance features for the two FFBs to make road features extracted from the two data sources guide each other so that the road areas in one data source are emphasized by the guidance features provided by the other data source, and so the input feature adds to the guided result though the residual learning process. The concatenation of two mutual guidance results serves as the fusion result (i.e., $C^{(i)}$), which is added to the feature maps of the corresponding layer in the decoder. In summary, through the AFM, we generate a comprehensive fusion feature that includes not only road features from two data sources but also the enhanced information about two kinds of road features.

C. Multitask Decoder Learning

According to the different characteristics of the two subtasks, we design two decoders to extract road regions and intersections. In the road decoder, inspired by the 1-D decoder block [14], which can leverage multiple rectangular convolution kernels for transposing convolutions to capture the information in transverse, longitudinal, oblique upper and oblique bottom directions and reducing the gaps in road prediction, here, we use four 1-D decoders to better recover the slender geometry of the road.

We propose an intersection decoder that not only captures special features of the intersection itself but also utilizes the learned information from the road decoder as the intersection obtained by the postprocessing of the road region. This is implemented with two guiding principles. On the one hand, the original decoder block of LinkNet [45] serves as intersection decoder blocks to preserve junction information. On the other hand, we add the skip features not only from the corresponding encoder block result but also from the corresponding road decoder block at each scale to improve the performance of the intersection detection task by using the learned road region information.

D. Adaptive Loss Function

In the field of road information detection, the proportion of road segmentation cases is relatively small, only approximately 10%. In the case of extreme imbalances between positive and negative samples, the effective information of a smaller positive sample becomes less pronounced after averaging, and the loss of information is dominated by a large number of negative samples. Therefore, it is necessary to integrate a balancing factor into the loss function to balance the uneven proportions of positive and negative samples. Moreover, because the background information ratio is too large, the probability of judging the confused pixels as the background is greatly increased, so the loss function should reduce the loss of easy-to-classify samples and pay more attention to the difficult and misclassified samples. To address the class imbalance problem, background information is erroneously divided. The focal loss, which can deal with category imbalances and determine the categories for confusing pixels, is introduced in this study and is defined as (2)

$$\begin{aligned} L_{fl} = & -\frac{1}{N}\sum_{i=1}^{N}(\alpha(1-y_i')\gamma y_i \log y_i' \\ & + (1-\alpha)\, y_i'^{\gamma}\,(1-y_i)\log(1-y_i')) \end{aligned} \tag{2}$$

where N is the number of pixels in the input image, y_i and y_i' are the ground truth and prediction of pixel i, respectively, and α is the balance factor to balance positive and negative samples, we set as 0.25. γ is the adjustment factor, and to adjust the rate of simple sample weight reduction, which is set as 2.

Therefore, we employ the focal loss function as our loss function to tackle same foreground issues. We minimize the focal loss of road detection and intersection extraction by gradient descent.

Let p_i' denote the ground truth of the ith pixel, p_i represents the probability that the ith pixel belongs to the road. Then, a focal loss function for the road can be defined as follows:

$$L_{\text{road}} = -\frac{1}{N}\sum_{i=1}^{N}(0.25(1-p_i')^2 p_i \log p_i' + 0.75 p_i'^2 (1-p_i)\log(1-p_i')). \tag{3}$$

Similarly, a focal loss function for the intersection can be defined as

$$L_{\text{intersection}} = -\frac{1}{N}\sum_{i=1}^{N}(0.25(1-q_i')^2 q_i \log q_i' + 0.75 q_i'^2 (1-q_i)\log(1-q_i')) \tag{4}$$

where, q_i and q_i' are the ground truth and prediction of intersection pixel i, respectively. Hence, the overall loss function of our network is summation of the weighted losses in (3) and (4), which can be described as (5):

$$\text{Loss} = \alpha L_{\text{road}} + (1-\alpha) L_{\text{intersection}} \tag{5}$$

where α and (1-α) are the weighting parameters.

Algorithm 1 presents the outline of the MTMSAFM network training process. We first construct the training instances from multisource data, which contain RS images and trajectories (lines 1–3). Then, MTMSAFM network is trained via backpropagation and Adam (lines 4–22).

Algorithm 1: Training MTMSAFM Network.

Input: Trajectory feature map: $\{\text{Tra}_0, \ldots, \text{Tra}_{n\text{-}1}\}$;
RS images: $\{\text{RS}_0, \ldots, \text{RS}_{n\text{-}1}\}$;
road label: $\{\text{L_road}_0, \ldots, \text{L_road}_{n\text{-}1}\}$;
Intersection label: $\{\text{L_intersection}_0, \ldots, \text{L_intersection}_{n\text{-}1}\}$;
output: MTMSAFM network model M
// construct training instance
1 $D \leftarrow \varnothing$
2 **for** $i = 0$ to $n-1$ **do**
3 put a training instance ($\{\text{Tra}_i, \text{RS}_i\}$, Road_l_i, Intersection_l_i) into D
// train the model
4 Initialize the parameters θ of M
5 **repeat**
6 randomly select a batch of instances D_b from D
//encoder block
7 **for** $l = 1$ to num(encoders) **do**
8 $F_R{}^{(l)} = \text{encoder_RS}^{(l)}(\{\text{RS}_0, \ldots \text{RS}_k\} \in D_b)$
9 $F_T{}^{(l)} = \text{encoder_trajectory}^{(l)}(\{\text{Tra}_0, \ldots \text{Tra}_k\} \in D_b)$
10 $C^{(l)} = \text{AFM}(F_R{}^{(l)}, F_T{}^{(l)})$
// center block
11 $D_{\text{road}}{}^{(\text{num(decoders)})} = \text{decoder_road}^{(\text{num(decoders)})}(\text{C}^{(\text{num(decoders)})})$
12 $D_{\text{jun}}{}^{(\text{num(decoders)})} = \text{decoder_intersection}^{(\text{num(decoders)})}(\text{C}^{(\text{num(decoders)})})$
// decoder block
13 **for** $m = 1$ to num(decoders) -1 **do**
14 $D_{\text{road}}{}^{(m)} = \text{decoder_road}^{(m)}(D\text{road}^{(m+1)}) + C^{(m)}$
15 $D_{\text{jun}}{}^{(m)} = \text{decoder_intersection}^{(m)}(D\text{jun}^{(m+1)}) + C^{(m)} + D\text{road}^{(m)}$
// prediction block
16 $\text{P_road} = \text{prediction_block}(D_{\text{road}}{}^{(1)})$
17 $\text{P_intersection} = \text{prediction_block}(D_{\text{jun}}{}^{(1)})$
//loss [equations (3)–(5)]
18) $\text{Loss}_{\text{road}} = \text{lossfunction}(\text{P_road}, \text{L_road})$
19 $\text{Loss}_{\text{intersection}} = \text{lossfunction}(\text{P_ intersection}, \text{L_ intersection})$
20 $\text{Loss} = \text{Loss}_{\text{road}} + \text{Loss}_{\text{intersection}}$
21 find θ by minimizing the Loss
22 **until** stopping criteria is met
23 output the learned MTMSAFM network model M.

IV. Experiments

To evaluate the effectiveness of the proposed framework, we conduct extensive experiments. The experiments consist of two procedures, including comparison with state-of-the-art image segmentation methods and testing of the effect of the deep learning network itself. All the deep learning networks are implemented by PyTorch and trained based on two NVIDIA Titan 1080ti GPUs.

A. Data

As far as we know, there is only one existing public dataset, named Porto dataset, which leverages both RS images and GPS trajectories for road extraction [16]. The Porto dataset contains RS image data with 1 m resolution and road ground truth image with 10 pixel width covering about 195 km^2, and it provides open-sourced trajectory dataset.[1] We also make two datasets by ourselves. One of them, named the Wuhan dataset, which covers approximately 175 km^2. The other dataset is Beijing dataset with the coverage of 88 km^2. All the RS images of the datasets are extracted from the Google Map API with resolutions of approximately 50 cm/pixel. For GPS trajectories, Wuhan trajectories are collected from approximately 2000 taxis in Wuhan from May 29, 2014 to June 7, 2014. The Beijing trajectories were generated by approximately 12 000 taxis over 14 days in 2012.

For the two datasets generated by ourselves, there are no labeled road maps and intersection maps. To generate ground truths for the images, we introduce the road network from

[1] [Online]. Available: www.kaggle.com/c/pkdd-15-predict-taxi-service-trajectory-i/data

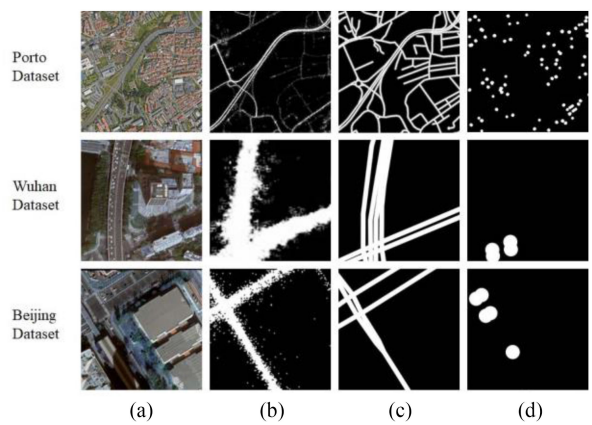

Fig. 5. Examples of samples in Porto dataset, Wuhan dataset, and the Beijing dataset. (a) RS images. (b) Trajectory feature maps. (c) Road ground truths. (d) Intersection ground truths.

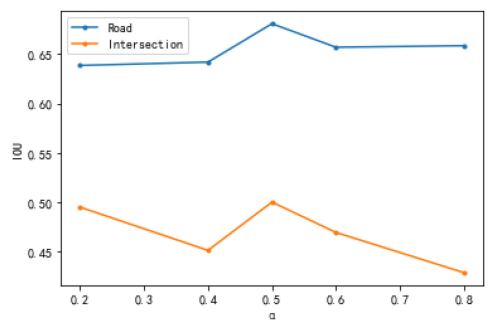

Fig. 6. Effect of weight parameter α.

OpenStreetMap[2] The selected road centerline consists of roads with a certain grade in the field named the "highway" of OSM. Meanwhile, the junction points of roads at the same elevation (the z_order field in OSM) are generated based on ArcMap. As our task is the image segmentation problem, we transform the road centerlines and junction points into images by rendering with fixed width. In detail, the road region labels are generated by marking 10-m widths, which are the average road widths, and the points are drawn with 30 m diameters, as some roads at high levels possess multiple lanes and a wider radius can cover the whole intersection. Examples of samples in three datasets are shown in Fig. 5.

For the trajectory feature map construction, we split the experimental region into $n \times n$ meter grids and calculate the density of trajectory points in each grid. We set n equal to 3 m and then resample the trajectory feature map to the same resolution as the raw RS image. For the Porto dataset with resolution 1 m, we randomly crop it to 256×256 samples without overlap. Considering that the higher resolution of the Wuhan dataset and Beijing dataset, these two datasets are clipped to 512×512 sizes without overlap for implementation. Finally, for Porto dataset, there are 2622 samples in the training dataset and 196 samples in the test dataset. 2867 images for the training region and 325 images for the test region are constructed for Wuhan dataset. The Beijing training dataset includes 2135 samples, and the test dataset consists of 165 samples.

[2][Online]. Available: www.openstreetmap.org

B. Implementation Details

We used the same experimental settings for all deep learning based models to minimize the influence of other factors. In detail, all the models were implemented by Pytorch framework and Adam algorithm was employed as our optimizer to train the network with a same initial learning rate ($2e-4$). We reduce the learning rate by five every three times the training loss decreases slowly.

We adopted the data augmentation methods to prevent overfitting. Specifically, the input images were first randomly scaled (from 0.9 to 1.1) to receive patches of the same size as the input. Subsequently, the patches were randomly flipped in horizontal and vertical directions, and then rotated with a step of 90°, and the above random threshold was all 0.5. By doing so, each input image was increased from 1 to $2 \times 2 \times 2 \times 2$ (16 times).

C. Metrics

We employ three well-known metrics [18] to evaluate the performance of the network for road region segmentation and intersection segmentation. These include completeness (COM), correctness (COR), and quality (Q), which are also called recall, precision, and intersection over union (IoU), respectively. COM is the ratio of the detected true pixels to ground true pixels. COR is the percentage of matched pixels out of the total number of segmentation pixels. Q is calculated by COR and COM. The three metrics can be computed as follows:

$$\mathrm{COM} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{6}$$

$$\mathrm{COR} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{7}$$

$$\mathrm{Q} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN} + \mathrm{FP}}. \tag{8}$$

Here, TP, FP, and FN represent the counts of true positives, false positives, and false negatives, respectively.

D. Multitask Weighted Parameter

To evaluate the effect of weight parameters on the multi task learning, we evaluated the model by setting the α weighted parameter from 0.2 to 0.8. A small α emphasis more on the task of intersection extraction, and α more concerned about road detection. The results are shown in Fig. 7. We found that when setting α as 0.5, proposed MTMSAF network achieves best performance on both road extraction and intersection detection tasks.

E. Segmentation Accuracy Comparison

To evaluate the effectiveness of road region detection and intersection extraction by the proposed MTMSAF network, we incorporate the following five state-of-the-art road extraction methods for comparison, which can be divided into three

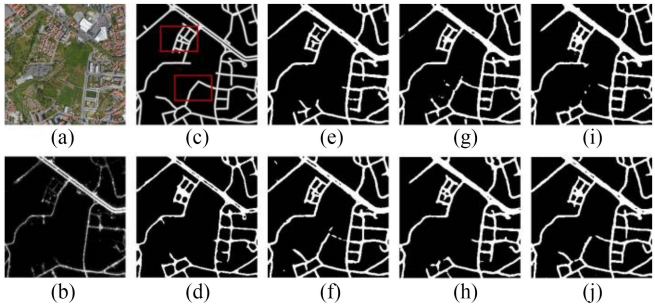

Fig. 7. Visual comparison of road extraction results based on Porto dataset by different methods. (a) RS image. (b) Trajectories. (c) Road ground truth. (d) DeepLabV3+. (e) LinkNet. (f)D-LinkNet. (g) LinkNet-1D. (h) MTMS network. (i) MSAF network. (j) MTMSAF network.

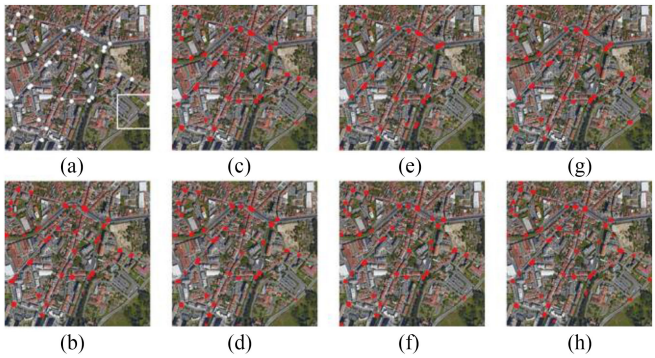

Fig. 8. Visual comparison of intersection extraction results based on Porto dataset by different methods. (a) RS image and intersection ground truth. (b) DeepLabV3+. (c) LinkNet. (d) D-LinkNet. (e) LinkNet-1D. (f) MTMS network. (g) MSAF network. (h) MTMSAF network.

categories. The first category is three outstanding image segmentation methods, including U-Net [29], LinkNet [45], and deeplabV3+[27]. The second category is road segmentation approach based on RS image contains D-LinkNet [43]; the third category is a fusion method for road information detection, called LinkNet-1D [14].

1) Porto Dataset: The first experiment is performed on the public Porto dataset to compare the performances of the road information detection methods. All the methods use the same data processing techniques and are evaluated on the same test dataset. The input of the five compared networks is the concatenation of the RS image and trajectory feature map with six channels. The MTMS network is the simplified version of the proposed MTMSAF network, which uses regular convolutions instead of PAConvs in FFB. MSAF network means only detect one task by fusion RS image and trajectory feature map, which has the same encoder with the MTMSAF network but only has one decoder stream, either for road detection or intersection extraction.

Fig. 7 demonstrates the road extraction results based on different deep neural networks. We can see from Fig. 7 that the main roads have been predicted completely, but for small roads covered by few trajectories, or roads c, the road detection performances vary. The red boxes represent the road extraction in the above two cases. The road detections based on all networks except the MTMSAF and LinkNet-1D network are missed. For road detection with complex geometry, only MSAF network and our proposed MTMSAF network successfully obtain continuous roads, as both of them contain AF module, which used to fuse in an effect way. Other compared methods easily result in discontinuities.

We visualize the road intersection extraction results by different methods in Fig. 8. Each point in red represents an intersection, and we overlay the intersection extraction results and road detection results on the RS image. From Fig. 11, we can see that our MSAF network and MTMSAF network perform best in the detection of the locations and number of intersections because we leverage AF module.

The quantitative comparison of the different methods for road information detection is listed in Table I. The columns represent the method names and three evaluation metrics, including the

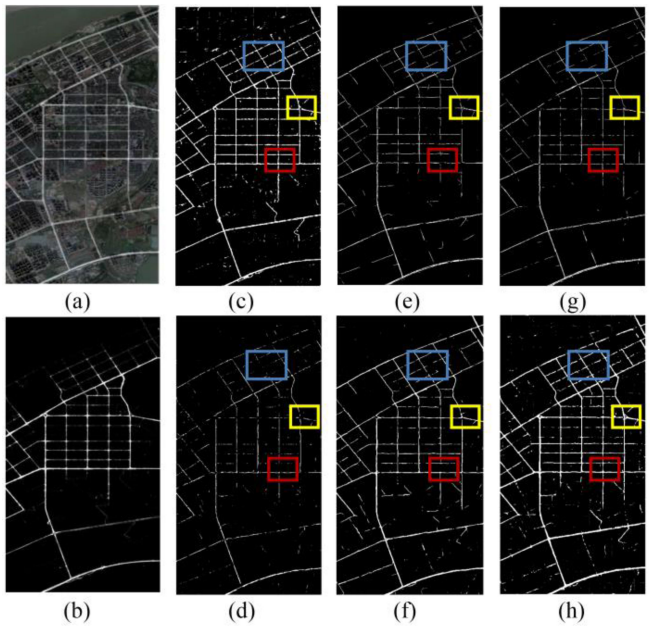

Fig. 9. Visual comparison of road extraction results based on Wuhan dataset I by different methods. (a) RS image and road ground truth. (b) Trajectories. (c) U-Net. (d) LinkNet. (e) D-LinkNet. (f) LinkNet-1D. (g) MTMS network. (h) MTMSAF network.

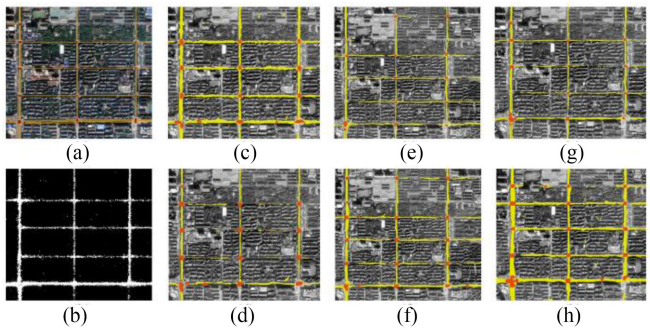

Fig. 10. Visual comparison of intersection extraction results based on Wuhan dataset I by different methods. (a) RS image and ground truth. (b) Trajectories. (c) U-Net. (d) LinkNet. (e) D-LinkNet. (f) LinkNet-1D. (g) MTMS network. (h) MTMSAF network.

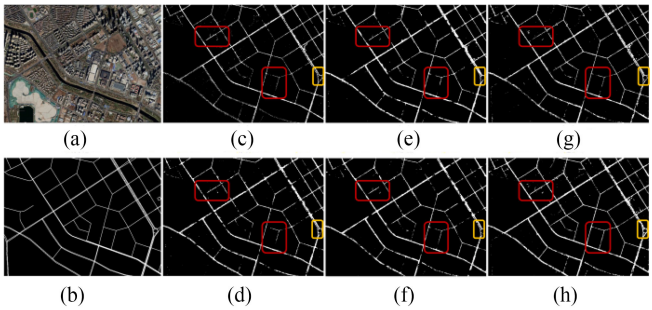

Fig. 11. Visual comparison of road detection results based on the Beijing dataset by different methods. (a) RS image. (b) Ground truth. (c) U-Net. (d) LinkNet. (e) D-LinkNet. (f) LinkNet-1D. (g) MTMS network. (h) MTMSAF network.

TABLE I
QUANTITATIVE COMPARISON OF DIFFERENT NETWORKS FOR ROAD INFORMATION DETECTION ON PORTO DATASET

Task	Road			Intersection		
Metric	**COM**	**COR**	**Q**	**COM**	**COR**	**Q**
DeepLabV3+	0.7799	0.7737	0.6328	0.5940	0.6275	0.4374
LinkNet	0.7807	0.7816	0.6355	0.5828	0.5999	0.4171
D-LinkNet	0.8090	0.7823	0.6558	0.6332	0.6439	0.4655
LinkNet-1D	0.7798	0.7830	0.6365	0.5595	0.5713	0.3881
MTMS network	0.7813	0.7920	0.6468	0.5985	0.6460	0.4486
MSAF network	0.8105	0.7966	0.6680	0.6294	0.5988	0.4375
MTMSAF network	**0.8106**	**0.8115**	**0.6809**	**0.6559**	**0.6817**	**0.4997**

average COM, COR, and Q for the whole test dataset. In each column, mark the best result in bold and the next best result in underline. Our proposed MTMSAF network achieves the highest values of all metrics, which is consistent with our visual evaluation. Among the compared methods, D-LinkNet is the best in road and intersection detection, and MTMSAF outperforms D-LinkNet by 2.51% and 3.42%.

2) Wuhan Dataset: Fig. 9 demonstrates the road extraction results based on different deep neural networks. Overall, the U-Net and MTMSAF networks show fewer road breaks than LinkNet, D-LinkNet, LinkNet 1D, and MTMS. Three square boxes are highlighted in Fig. 9 to exhibit the regional road extraction performances. In the blue box region, the U-Net and MTMSAF networks can generally extract some small roads, while other networks may have apparent missing roads or breaks. In the yellow box region, MTMSAF can reproduce the geometric shapes of complex crossings by fully integrating RS image information, and it does so much better than other networks. In the red box region, MTMSAF shows ideal road extraction results with good completeness and continuity, while U-Net may suffer from underlying breaks near road intersections, suggesting the benefit of MTMSAF after including the AFM. The MTMS network without an AFM demonstrates equivalent or even better performance than LinkNet and D-LinkNet spatially, while the MTMSAF network indicates obvious added values relative to MTMS in terms of road completeness and continuity.

Table II reports the results evaluated by the COM, COR, and Q metrics. From the table we can see, two of best results belong to the proposed MTMSAF network. The second highest value of Q belongs to the proposed MTMS network, which demonstrates that the connection between road detection and intersection extraction helps with road detection. For the COM value, our proposed MTMSAF network achieves the highest score, which verifies the validity of our proposed network. D-LinkNet possesses the highest value of COR because it detects the shortest total road length.

TABLE II
QUANTITATIVE COMPARISON OF DIFFERENT NETWORKS FOR ROAD REGION DETECTION ON WUHAN DATASET

Method	COM	COR	Q
U-Net	0.7085	0.5424	0.4339
LinkNet	0.3020	0.6605	0.2537
D-LinkNet	0.4329	**0.6645**	0.3544
LinkNet-1D	0.5807	0.5786	0.4042
MTMS network	0.6951	0.5709	0.4409
MSAF network (road detection)	0.5103	0.6151	0.3847
MTMSAF network	**0.7329**	0. 5645	**0.4566**

TABLE III
QUANTITATIVE COMPARISON OF DIFFERENT NETWORKS FOR INTERSECTION EXTRACTION ON WUHAN DATASET I

Method	COM	COR	Q
U-Net	0.5477	0.5881	0.3712
LinkNet	0.6374	0.5683	0.4173
D-LinkNet	**0.7128**	0.5437	0.4353
LinkNet-1D	0.6184	**0.6217**	0.4335
MTMS network	0.5660	0.5650	0.3874
MSAF network (intersection extraction)	0.5667	0.6155	0.4005
MTMSAF network	0.6392	0.6155	**0.4374**

For intersection extraction, we also compare the above algorithms. For U-Net, LinkNet, D-LinkNet, and LinkNet 1D, we use the concatenation of RS images and trajectories to train an intersection extraction network and then apply the trained model to the test data. Fig. 10 shows the qualitative comparison of the intersection extraction results. The intersection extraction results are overlaid on the road detection results to verify the improvement in each of the two tasks in our proposed network. For U-Net, LinkNet, D-LinkNet, and LinkNet 1D, the intersection extraction results are inconsistent with the junction of the road detection, and there are some missing and false extractions, as shown in the red box in Fig. 10. However, for the MTMS network and MTMSAF network, the locations of the extracted intersections are exactly the crossing of the road detection, which demonstrates that multitask learning framework can leverage the connections between roads and intersections. Moreover, the unlabeled but true intersections as well as multiple intersections at large scales are detected by MTMSAF, which suggests that MTMSAF can capture the fused information well with the leveraging of the AFM.

Table III reports the quantitative comparison of intersection extraction results based on the Wuhan dataset. D-LinkNet achieves the highest COM value, followed by D-LinkNet and MTMSAF. This means that D-LinkNet detects fewer false negatives and the low COR value in D-LinkNet suggests a high false alarm rate of true positives. LinkNet-1D possesses the highest COR value, followed by MTMSAF. The MTMSAF network exhibits the highest Q score, suggesting an overall better

TABLE IV
QUANTITATIVE COMPARISON OF DIFFERENT NETWORKS FOR ROAD REGION DETECTION ON THE BEIJING DATASET

Method	COM	COR	Q
U-Net	0.5771	0.6747	0.4360
LinkNet	0.6532	**0.6808**	0.4887
D-LinkNet	0.6863	0.6616	0.4883
LinkNet-1D	0.6543	0.6723	0.4886
MTMS network	0.6872	0.6690	0.4906
MTMSAF network	**0.7346**	0.6340	**0.4936**

TABLE V
QUANTITATIVE COMPARISON OF DIFFERENT NETWORKS FOR INTERSECTION EXTRACTION ON THE BEIJING DATASET

Method	COM	COR	Q
U-Net	0.1492	0.472	0.1286
LinkNet	0.3415	0.6773	0.2707
D-LinkNet	0.3767	0.4652	0.2669
LinkNet-1D	0.3178	0.6152	0.2622
MTMS network	0.3950	0.7825	0.3344
MTMSAF network	**0.4182**	**0.8145**	**0.3688**

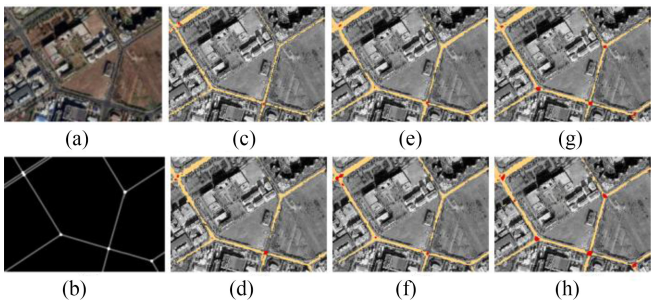

Fig. 12. Visual comparison of intersection extraction results based on the Beijing dataset by different methods. (a) RS image. (b) Ground truth. (c) U-Net. (d) LinkNet. (e) D-LinkNet. (f) LinkNet-1D. (g) MTMS network. (h) MTMSAF network.

performance than other methods, resulting from the relatively high COM and high COR values.

3) Beijing Dataset: This experiment is performed on the Beijing dataset, and its ground truth is also generated from OSM. Fig. 11 represents the road extraction results of different methods for the whole test area. U-Net, LinkNet and the variants of LinkNet, such as D-LinkNet and LinkNet 1D possess many discontinuities in their road prediction results. Among the compared methods, our MTMSAF network obtains the best results not only in the completeness of the small road shown in the red box but also in the segmentation of detailed information shown in the yellow box. This is attributed to our proposed AFM, which can better capture the enhanced information of RS images and trajectories. Table V lists the results evaluated by the COM, COR, and Q metrics. Our proposed MTMS network and MTMSAF network leverage the road detection task and intersection extraction task simultaneously. The extracted intersection helps with road connectivity around the intersection, so these methods obtain the best COM values, especially the MTMSAF network. Moreover, the MTMSAF network performs better in terms of Q. However, for the COR metric, the baseline LinkNet and U-Net perform best. However, their performances in terms of COM are relatively low, which means that these methods can reject correctly and are closer to the ground truth. It should be noted that the ground truth is not very accurate, and the detected roads exist in reality but are not included in the ground truth, which leads to low COR values for our methods.

The road intersection extraction results obtained by the different methods in a subset testing region of the Beijing dataset are exhibited in Fig. 12. From Fig. 12, we can see that our MTMS network and MTMSAF network perform best; they can extract all intersections, but neither of the comparative methods can detect all intersections. Table V represents the quantitative evaluation measured in terms of COM, COR, and Q. In the table, the proposed two methods outperform other networks in terms of COM, COR, and Q. The low Q values obtained by nearly all the networks are likely due to the low quality of intersection labels for OSM because of the complex transportation conditions in the Beijing region.

V. CONCLUSION

In this article, we propose a road network generation framework based on RS images and trajectories. The framework can fuse road features extracted from two data sources at different levels with a guidance strategy to obtain enhanced information from the two data sources and predict road regions and intersection tasks at the same time. Experiments show that the method can pay more attention to detecting continuous, complete and detailed road information based on the fusion of the enhanced information from RS images and trajectories. Compared to other methods, our method produces less discontinuity between roads and intersections and fewer false positive intersections due to the simultaneous combination of road detection and intersection extraction tasks. The proposed network improves the integrity of the road with the AFM. In the future, we plan to extract navigable information such as turning restrictions to enrich our map. Accurate ground truths play an important role in prediction performance. Relying solely on OSM yields limited improvement in road and intersection extraction performances. In a future study, we will focus on how to build ground truth labels in an accurate and efficient way.

REFERENCES

[1] X. Du and K. K. Tan, "Comprehensive and practical vision system for self-driving vehicle lane-level localization," *IEEE Trans. Image Process.,* vol. 25, no. 5, pp. 2075–2088, May 2016.

[2] C. E. Soulard, W. Acevedo, and S. V. Stehman, "Removing rural roads from the National Land Cover Database to create improved urban maps for the United States, 1992 to 2011," *Photogrammetric Eng. Remote Sens.,* vol. 84, no. 2, pp. 101–109, 2018.

[3] C. Zhang, Y. Li, L. Xiang, F. Jiao, and S. Li, "Generating road networks for old downtown areas based on crowd-sourced vehicle trajectories," *Sensors,* vol. 21, no. 1, 2021, Art. no. 235.

[4] S. Karagiorgou, D. Pfoser, and D. Skoutas, "A layered approach for more robust generation of road network maps from vehicle tracking data," *ACM Trans. Spatial Algorithms Syst.,* vol. 3, no. 1, pp. 1–21, 2017.

[5] J. Li, Q. Qin, J. Han, L.-A. Tang, and K. H. Lei, "Mining trajectory data and geotagged data in social media for road map inference," *Trans. GIS,* vol. 19, no. 1, pp. 1–18, 2014.

[6] M. Zhou, H. Sui, S. Chen, J. Wang, and X. Chen, "BT-RoadNet: A boundary and topologically-aware neural network for road extraction from high-resolution remote sensing imagery," *ISPRS J. Photogrammetry Remote Sens.,* vol. 168, pp. 288–306, 2020.

[7] Y. Wei, Z. Wang, and M. Xu, "Road structure refined CNN for road extraction in aerial image," *IEEE Geosci. Remote Sens. Lett.,* vol. 14, no. 5, pp. 709–713, May 2017.

[8] R. Alshehhi and P. R. Marpu, "Hierarchical graph-based segmentation for extracting road networks from high-resolution satellite images," *ISPRS J. Photogrammetry Remote Sens.,* vol. 126, pp. 245–260, 2017.

[9] S. Xu, R. Wang, and H. Zheng, "Road curb extraction from mobile LiDAR point clouds," *IEEE Trans. Geosci. Remote Sens.,* vol. 55, no. 2, pp. 996–1009, Feb. 2017.

[10] W. Shi, S. Shen, and Y. Liu, "Automatic generation of road network map from massive GPS, vehicle trajectories," in *Proc. 12th Int. IEEE Conf. Intell. Transp. Syst.*, 2009, pp. 1–6. [Online]. Available: http://dx.doi.org/10.1109/itsc.2009.5309871

[11] R. Alshehhi, P. R. Marpu, W. L. Woon, and M. D. Mura, "Simultaneous extraction of roads and buildings in remote sensing imagery with convolutional neural networks," *ISPRS J. Photogrammetry Remote Sens.,* vol. 130, pp. 139–149, 2017.

[12] C. Chen, C. Lu, Q. Huang, Q. Yang, D. Gunopulos, and L. Guibas, "City-scale map creation and updating using GPS collections," presented at the *Proc. 22nd ACM SIGKDD Int. Conf. Knowl. Discov. Data Mining*, San Francisco, California, USA, 2016, pp. 1465–1474. [Online]. Available: http://dx.doi.org/10.1145/2939672.2939833

[13] M. Deng *et al.*, "Generating urban road intersection models from low-frequency GPS trajectory data," *Int. J. Geographical Inf. Sci.,* vol. 32, no. 12, pp. 2337–2361, 2018.

[14] T. Sun, Z. Di, P. Che, C. Liu, and Y. Wang, "Leveraging crowdsourced GPS data for road extraction from aerial imagery," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 7501–7510. [Online]. Available: http://dx.doi.org/10.1109/cvpr.2019.00769

[15] Y. Li, L. Xiang, C. Zhang, and H. Wu, "Fusing taxi trajectories and RS images to build road map via DCNN," *IEEE Access,* vol. 7, pp. 161487–161498, 2019.

[16] H. Wu, H. Zhang, X. Zhang, W. Sun, B. Zheng, and Y. Jiang, "DeepDualMapper: A gated fusion network for automatic map extraction using aerial images and trajectories," in *Proc. AAAI Conf. Artif. Intell.,* 2020, vol. 34, no. 1, pp. 1037–1045.

[17] C. Zhang, L. Xiang, S. Li, and D. Wang, "An intersection-first approach for road network generation from crowd-sourced vehicle trajectories," *ISPRS Int. J. Geo-Inf.,* vol. 8, no. 11,2019, Art. no. 473.

[18] G. Cheng, Y. Wang, S. Xu, H. Wang, S. Xiang, and C. Pan, "Automatic road detection and centerline extraction via cascaded end-to-end convolutional neural network," *IEEE Trans. Geosci. Remote Sens.,* vol. 55, no. 6, pp. 3322–3337, Jun. 2017.

[19] G. Cheng, F. Zhu, S. Xiang, Y. Wang, and C. Pan, "Accurate urban road centerline extraction from VHR imagery via multiscale segmentation and tensor voting," *Neurocomputing,* vol. 205, pp. 407–420, 2016.

[20] X. Lu *et al.*, "Multi-scale and multi-task deep learning framework for automatic road extraction," *IEEE Trans. Geosci. Remote Sens.,* vol. 57, no. 11, pp. 9362–9377, Nov. 2019.

[21] Y. Liu, J. Yao, X. Lu, M. Xia, X. Wang, and Y. Liu, "RoadNet: Learning to comprehensively analyze road networks in complex urban scenes from high-resolution remotely sensed images," *IEEE Trans. Geosci. Remote Sens.,* vol. 57, no. 4, pp. 2043–2056, Apr. 2019.

[22] X. Yang, X. Li, Y. Ye, R. Y. K. Lau, X. Zhang, and X. Huang, "Road detection and centerline extraction via deep recurrent convolutional neural network U-Net," *IEEE Trans. Geosci. Remote Sens.,* vol. 57, no. 9, pp. 7209–7220, Sep. 2019.

[23] A. Gholamreza, "A new statistical-based kurtosis wavelet energy feature for texture recognition of SAR images," *IEEE Trans. Geosci. Remote Sens.,* vol. 50, no. 11, pp. 4358–4368, Nov. 2012.

[24] Z. Tirandaz and G. Akbarizadeh, "A two-phase algorithm based on kurtosis curvelet energy and unsupervised spectral regression for segmentation of SAR images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.,* vol. 9, no. 3, pp. 1244–1264, Mar. 2016.

[25] M. Zalpour, G. Akbarizadeh, and N. Alaei-Sheini, "A new approach for oil tank detection using deep learning features with control false alarm rate in high-resolution satellite imagery," *Int. J. Remote Sens.* vol. 41, no. 6, pp. 2239–2262, 2020.

[26] N. Davari, G. Akbarizadeh, and E. Mashhour, "Intelligent diagnosis of incipient fault in power distribution lines based on corona detection in UV-Visible videos," *IEEE Trans. Power Del.,* to be published, doi: 10.1109/TPWRD.2020.3046161.

[27] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder-decoder with atrous separable convolution for semantic image segmentation," in *Computer Vision–ECCV*. New York, NY, USA: Springer International Publishing, 2018, pp. 833–851.

[28] V. Mnih and G. E. Hinton, "Learning to detect roads in high-resolution aerial images," in *Computer Vision–ECCV 2010*. Berlin, Germany: Springer, 2010, pp. 210–223.

[29] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Lecture Notes in Computer Science*. New York, NY, USA: Springer, 2015, pp. 234–241.

[30] Z. Zhang, Q. Liu, and Y. Wang, "Road extraction by deep residual U-Net," *IEEE Geosci. Remote Sens. Lett.,* vol. 15, no. 5, pp. 749–753, May 2018.

[31] G. Cheng *et al.*, "Recognizing road from satellite images by structured neural network," *Neurocomputing,* vol. 356, pp. 131–141, 2019.

[32] C. Tao, J. Qi, Y. Li, H. Wang, and H. Li, "Spatial information inference net: Road extraction using road-specific contextual information," *ISPRS J. Photogrammetry Remote Sens.,* vol. 158, pp. 155–166, 2019.

[33] R. Stanojevic, S. Abbar, S. Thirumuruganathan, S. Chawla, F. Filali, and A. Aleimat, "Robust road map inference through network alignment of trajectories," in *Proc. SIAM Int. Conf. Data Mining*, 2018, pp. 135–143.

[34] J. Qiu and R. Wang, "Automatic extraction of road networks from GPS traces," *Photogrammetric Eng. Remote Sens.,* vol. 82, no. 8, pp. 593–604, 2016.

[35] L. Cao and J. Krumm, "From GPS traces to a routable road map," in presented at the *Proc. 17th ACM SIGSPATIAL Int. Conf. Adv. Geographic Inform. Syst.*, Seattle, Washington, 2009, pp. 3–12. [Online]. Available: http://dx.doi.org/10.1145/1653771.1653776

[36] L. Tang, C. Ren, Z. Liu, and Q. Li, "A road map refinement method using delaunay triangulation for big trace data," *ISPRS Int. J. Geo-Inf.,* vol. 6, no. 2, 2017, Art. no. 45.

[37] J. Biagioni and J. Eriksson, "Map inference in the face of noise and disparity," in presented at the *Proc. 20th Int. Conf. Adv. Geographic Inform. Syst.*, 2012, pp. 79–88. [Online]. Available: http://dx.doi.org/10.1145/2424321.2424333

[38] R. Zheng, Q. Liu, W. Rao, M. Yuan, J. Zeng, and Z. Jin, "Topic model-based road network inference from massive trajectories," in *Proc. 18th IEEE Int. Conf. Mobile Data Manage. (MDM)*, 2017, pp. 246–255. [Online]. Available: http://dx.doi.org/10.1109/mdm.2017.41

[39] S. Zourlidou and M. Sester, "Intersection detection based on qualitative spatial reasoning on stopping point clusters," *Int. Arch. Photogrammetry, Remote Sens. Spatial Inf. Sci.-ISPRS Arch.*, vol. 41, pp. 269–276, 2016.

[40] R. Mariescu-Istodor and P. Fränti, "CellNet: Inferring road networks from GPS trajectories," *ACM Trans. Spatial Algorithms Syst. (TSAS),* vol. 4, no. 3, pp. 1–22, 2018.

[41] J. Yuan and A. M. Cheriyadat, "Image feature based GPS trace filtering for road network generation and road segmentation," *Mach. Vis. Appl.,* vol. 27, no. 1, pp. 1–12, 2015.

[42] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR),* 2016, pp. 770–778. [Online]. Available: http://dx.doi.org/10.1109/cvpr.2016.90

[43] L. Zhou, C. Zhang, and M. Wu, "D-LinkNet: Linknet with pretrained encoder and dilated convolution for high resolution satellite imagery road extraction," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW),* 2018, pp. 192–1924. [Online]. Available: http://dx.doi.org/10.1109/cvprw.2018.00034

[44] H. Su, V. Jampani, D. Sun, O. Gallo, E. Learned-Miller, and J. Kautz, "Pixel-adaptive convolutional neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2019, pp. 11158–11167. [Online]. Available: http://dx.doi.org/10.1109/cvpr.2019.01142

[45] A. Chaurasia and E. Culurciello, "LinkNet: Exploiting encoder representations for efficient semantic segmentation," in *Proc. IEEE Vis. Commun. Image Process. (VCIP)*, 2017, pp. 1–4. [Online]. Available: http://dx.doi.org/10.1109/vcip.2017.8305148

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